

Relationship Types & Relationship Sets

Q. Explain the concepts of Relationship Types and Relationship Sets. Discuss the Degree of Relationship and Mapping Cardinalities (1:1, 1:N, M:N) with examples.

Definition to Remember

A **Relationship** diagram moment

1. Relationship Type and Relationship Set

* **Relationship Type:** Association between entity types. E.g., `WORKS\_FOR` links `EMPLOYEE` and `DEPARTMENT`.

* **Relationship Set**

2. Degree of Relationship

The Degree = number of entity types participating in the relationship.

Degree	Name	Example
1	Unary	EMPLOYEE
2	Binary	WORKS_FOR
3	Ternary	EMPLOYEE, DEPARTMENT, WORKS_FOR

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3. Mapping Cardinalities

Mapping Cardinality defines the maximum number of relationship instances an entity can participate in.

1. One-to-One (1:1)

* One

2. One-to-Many (1:N)

* One

3. Many-to-One (N:1)

* Multi

4. Many-to-Many (M:N)

* Entit

1:1 -> A <R> B

1:N ->

4. Participation Constraints

Type	Description	ER Symbol
1:1	One-to-One	Two entity types connected by a line with a double bar at each end.
1:N	One-to-Many	One entity type connected to another by a line with a double bar at the one end and a crow's foot notation at the many end.
M:N	Many-to-Many	Two entity types connected by a line with a double bar at each end and a crow's foot notation in the middle.

* **Total Example:** Every `EMPLOYEE` must belong to a `DEPARTMENT`.

* **Pa**

Must-Write Points (for 10 marks)

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Quick Recall

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